Module -IV

Z-TRANSFORMS

Definition

Z-transform

If the function $f(n)$ is defined for $n = 0,1,2,\ldots$ and $f(n) = 0$ *for* $n < 0$; then $f(0), f(1), f(2), \ldots$ is a sequence, denoted by $\{f(n)\}$. The Z-transform of the sequence $\{f(n)\}$ is defined as

$$F(z) = Z[f(n)] = \sum_{n=0}^{\infty} f(n)z^{-n}, \text{if the series converges.}$$

Where z is a complex number. This Z-transform is called one-sided or unilateral Z-transform.

Definition

If the function $f(n)$ is defined for $n = 0, \pm 1, \pm 2, \ldots$, then Z-transform of the sequence $\{f(n)\}$ is defined as

$$F(z) = Z[f(n)] = \sum_{n=-\infty}^{\infty} f(n)z^{-n}.$$

This Z-transform is called two-sided or bilateral Z-transform.

Definition

If the continuous function $f(t)$ is defined for the sampled values $t = nT, n = 0,1,2,\ldots$, then the Z-transform of $\{f(t)\}$ is defined as

$$F(z) = Z[f(nt)] = \sum_{n=0}^{\infty} f(nT)z^{-n}$$

Region of Convergence

The value of z for which Z-Transform converges is called region of convergence.

Results:

1. $(1-x)^{-1} = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1$

2. $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots, \quad |x| < 1$

3. $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \cdots, \quad |x| < 1$

4. $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \cdots, \quad |x| < 1$

5. $e^{i\theta} = cos\,\theta + i\,sin\,\theta$

6. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

7. $n! = n(n-1)(n-2)\ldots 3.2.1 \quad and \quad 0! = 1$

8. $log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

9. $log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$

10. In the Geometric Progression $1 + r + r^2 + \cdots + r^{n-1}$

$$S_n = \left(\frac{1-r^n}{1-r}\right); r < 1$$

$$= \left(\frac{r^n - 1}{r-1}\right); r > 1 \qquad \text{(Or)}$$

In the Geometric Progression $1 + r + r^2 + \cdots + r^n$

$$S_{n+1} = \left(\frac{1-r^{n+1}}{1-r}\right); r < 1$$

$$= \left(\frac{r^{n+1} - 1}{r-1}\right); r > 1 \quad \text{where } r = \frac{t_2}{t_1}$$

Linearity Property of Z-transform:

The Z-transform is a linear transformation.

i.e., $Z[af(n) + bg(n)] = aZ[f(n)] + bZ[g(n)]$ Where a, b are any constants.

Z-Transform of Elementary Functions:

1. $Z[k] = \frac{kz}{z-1}$, k is a constant, $|z| > 1$

Given $f(n) = k$

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n)z^{-n}$$

$$Z[k] = \sum_{n=0} k\, z^{-n} = k \sum_{n=0} \frac{1}{z^n}$$

$$= k\left[1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots\right]$$

$$= k\left[1 - \frac{1}{z}\right]^{-1} \; if \; |z| > 1 \Rightarrow \frac{1}{|z|} < 1 \Rightarrow \left|\frac{1}{z}\right| < 1$$

$$= k\left[\frac{z-1}{z}\right]^{-1} = k\left[\frac{z}{z-1}\right]$$

$$\therefore Z[k] = \frac{kz}{z-1}$$

Note:

(i) $Z[1] = \frac{z}{z-1}$ (ii) $Z[2] = \frac{2\,z}{z-1}$

2. $Z[a^n] = \frac{z}{z-a}$, $|z| > |a|$

Given $f(n) = a^n$

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n) z^{-n}$$

$$Z[a^n] = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} a^n \frac{1}{z^n} = \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n$$

$$= \left[1 + \frac{a}{z} + \left(\frac{a}{z}\right)^2 + \left(\frac{a}{z}\right)^3 + \cdots\right]$$

$$= \left[1 - \frac{a}{z}\right]^{-1} \; if \; |z| > |a| \Rightarrow \frac{1}{|z|} < \frac{1}{|a|} \Rightarrow \left|\frac{a}{z}\right| < 1$$

$$= \left[\frac{z-a}{z}\right]^{-1}$$

$$Z[a^n] = \frac{z}{z-a}$$

Note:

(i) If $a = 1$ then $Z[(1)^n] = \frac{z}{z-1}$

(ii) If $a = -1$ then $Z[(-1)^n] = \frac{z}{z+1}$

(iii) $Z[e^{an}] = Z[(e^a)^n] = \frac{z}{z-e^a}$

3. $Z[n] = \frac{z}{(z-1)^2}$

Given $f(n) = n$

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n) z^{-n}$$

$$Z[n] = \sum_{n=0}^{\infty} n\, z^{-n} = \sum_{n=0}^{\infty} n \frac{1}{z^n} = \sum_{n=0}^{\infty} n \left(\frac{1}{z}\right)^n$$

$$= \left[0 + 1.\left(\frac{1}{z}\right) + 2.\left(\frac{1}{z}\right)^2 + 3.\left(\frac{1}{z}\right)^3 + \cdots\right]$$

$$= \frac{1}{z}\left[1 + 2.\left(\frac{1}{z}\right) + 3.\left(\frac{1}{z}\right)^2 + \cdots\right]$$

$$= \frac{1}{z}\left[1 - \frac{1}{z}\right]^{-2} = \frac{1}{z}\left[\frac{z-1}{z}\right]^{-2} = \frac{1}{z}\left[\frac{z}{z-1}\right]^2 = \frac{1}{z}\frac{z^2}{(z-1)^2}$$

$$\therefore Z[n] = \frac{z}{(z-1)^2}$$

4. $Z\left[\frac{1}{n}\right] = log_e\left(\frac{z}{z-1}\right), n > 0$

Given $f(n) = \frac{1}{n}$

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n) z^{-n}$$

$$Z\left[\frac{1}{n}\right] = \sum_{n=1}^{\infty} \frac{1}{n} z^{-n} = \sum_{n=1}^{\infty} \frac{1}{n} \frac{1}{z^n} = \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{1}{z}\right)^n$$

$$= \sum_{n=1}^{\infty} \frac{\left(\frac{1}{z}\right)^n}{n}$$

$$= \frac{\left(\frac{1}{z}\right)}{1} + \frac{\left(\frac{1}{z}\right)}{2} + \frac{\left(\frac{1}{z}\right)}{3} + \cdots$$

$$= -log\left(1 - \frac{1}{z}\right) = -log\left(\frac{z-1}{z}\right) = log\left(\frac{z-1}{z}\right)^{-1}$$

$$\therefore Z\left[\frac{1}{n}\right] = log\left(\frac{z}{z-1}\right)$$

5. $Z\left[\frac{1}{n!}\right] = e^{\frac{1}{z}}$

Given $f(n) = \frac{1}{n!}$

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n) z^{-n}$$

$$Z\left[\frac{1}{n!}\right] = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{1}{z^n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{z}\right)^n = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{z}\right)^n}{n!}$$

$$= 1 + \frac{\left(\frac{1}{z}\right)}{1!} + \frac{\left(\frac{1}{z}\right)^2}{2!} + \frac{\left(\frac{1}{z}\right)^3}{3!} + \cdots$$

$$Z\left[\frac{1}{n!}\right] = e^{\frac{1}{z}}$$

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6. $Z\left[\frac{a^n}{n!}\right] = e^{\frac{a}{z}}$

Given $f(n) = \frac{a^n}{n!}$

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n) z^{-n}$$

$$Z\left[\frac{a^n}{n!}\right] = \sum_{n=0}^{\infty} \frac{a^n}{n!} z^{-n} = \sum_{n=0}^{\infty} \frac{a^n}{n!} \frac{1}{z^n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{a}{z}\right)^n = \sum_{n=0}^{\infty} \frac{\left(\frac{a}{z}\right)}{n!}$$

$$= 1 + \frac{\left(\frac{a}{z}\right)}{1!} + \frac{\left(\frac{a}{z}\right)^2}{2!} + \frac{\left(\frac{a}{z}\right)^3}{3!} + \cdots$$

$$Z\left[\frac{a^n}{n!}\right] = e^{\frac{a}{z}}$$

Table of Z-Transforms:

Sl. No.	$f(n)$	$Z\{f(n)\} = F(z)$
1.	k, *a constant*	$\frac{kz}{z-1}$
2.	a^n	$\frac{z}{z-a}$
3.	$(-1)^n$	$\frac{z}{z+1}$
4.	n	$\frac{z}{(z-1)^2}$
5.	n^2	$\frac{z^2+z}{(z-1)^3}$
6.	n^3	$\frac{z^3+4z^2+z}{(z-1)^4}$
7.	na^n	$\frac{az}{(z-a)^2}$
8.	$\frac{1}{n}$	$log\left[\frac{z}{z-1}\right]$
9.	$\frac{1}{n+1}$	$zlog\left[\frac{z}{z-1}\right]$
10.	$\frac{a^n}{n!}$	$e^{\frac{a}{z}}$
11.	$r^n cosn\theta$	$\frac{z(z-rcos\theta)}{z^2-2zrcos\theta+r^2}$
12.	$r^n sinn\theta$	$\frac{zr\,sin\theta}{z^2-2zrcos\theta+r^2}$

13.	$cos\frac{n\pi}{2}$	$\frac{z^2}{z^2+1}$
14.	$sin\frac{n\pi}{2}$	$\frac{z}{z^2+1}$

Multiplication by n (or) Differentiation in the Z-domain

If $Z[f(n)] = F(z)$, then $Z[nf(n)] = -z\frac{d}{dz}F(z)$

Proof:

$$F(z) = Z[f(n)] = \sum_{n=0}^{\infty} f(n)z^{-n}$$

Differentiate w.r.to z,

$$\frac{d}{dz}F(z) = \sum_{n=0}^{\infty} f(n)(-n)z^{-n-1} = -\sum_{n=0}^{\infty} f(n)nz^{-n}z^{-1}$$

$$= -z^{-1}\sum_{n=0}^{\infty}\{nf(n)\}z^{-n}$$

$$= -\frac{1}{z}\sum_{n=0}^{\infty}\{nf(n)\}z^{-n}$$

$$\frac{d}{dz}F(z) = -\frac{1}{z}Z[nf(n)]$$

$$Z[nf(n)] = -z\frac{d}{dz}F(z)$$

Note:

In general, $Z[n^p f(n)] = (-z)^p\frac{d^p}{dz^p}F(z)$

Time Shifting Property

Shifting to the Right (First Shifting Theorem)

If $Z[f(n)] = F(z)$, then $Z[f(n-k)] = z^{-k}F(z) for\ k > 0$

Proof:

Given, $Z\{f(n)\} = F(z)$

We know that, $Z\{f(n)\} = \sum_{n=0}^{\infty} f(n)z^{-n}$

$$Z\{f(n-k)\} = \sum_{n=0}^{\infty} f(n-k)z^{-n}$$

Put $m = n-k$

Hence, $n = m+k$

If $n \to 0\ then\ m \to -k$

If $n \to \infty\ then\ m \to \infty$

$$Z\{f(n-k)\} = \sum_{m=-k}^{\infty} f(m)z^{-(m+k)}$$

$$= \sum_{m=-k}^{\infty} f(m)z^{-m}z^{-k}$$

$$Z\{f(n-k)\} = z^{-k}\sum_{m=-k}^{\infty} f(m)z^{-m}$$

But $f(m) = 0$ when $m < 0$, since we are considering one sided $Z-$ Transform

$$Z\{f(n-k)\} = z^{-k}\sum_{m=0}^{\infty} f(m)z^{-m}$$

$$= z^{-k}Z\{f(m)\}$$

$$Z\{f(n-k)\} = z^{-k}F(z)$$

Shifting to the Left (Second Shifting Theorem)

If $Z[f(n)] = F(z)$, then

$$Z\{f(n+k)\} = z^k\left[F(z) - f(0) - \frac{f(1)}{z} - \frac{f(2)}{z^2} - \cdots - \frac{f(k-1)}{z^{k-1}}\right], k > 0$$

Proof:

Given, $Z\{f(n)\} = F(z)$

We know that, $Z\{f(n)\} = \sum_{n=0} f(n)z^{-n}$

$$Z\{f(n+k)\} = \sum_{n=0}^{\infty} f(n+k)z^{-n}$$

Put $m = n + k$

Hence, $n = m - k$

If $n \to 0$ then $m \to k$

If $n \to \infty$ then $m \to \infty$

$$Z\{f(n+k)\} = \sum_{m=k}^{\infty} f(m)z^{-(m-k)} = \sum_{m=k}^{\infty} f(m)z^{-m+k}$$

$$= \sum_{m=k}^{\infty} f(m)z^{-m}z^{k}$$

$$= z^{k}\left\{\sum_{m=0}^{k-1} f(m)z^{-m} + \sum_{m=k}^{\infty} f(m)z^{-m} - \sum_{m=0}^{k-1} f(m)z^{-m}\right\}$$

$$= z^{k}\left\{\sum_{m=0}^{\infty} f(m)z^{-m} - \sum_{m=0}^{k-1} \frac{f(m)}{z^{m}}\right\}$$

$$= z^{k}\left\{Z\{f(n)\} - \left[f(0) + \frac{f(1)}{z} + \frac{f(2)}{z^{2}} + \cdots + \frac{f(k-1)}{z^{k-1}}\right]\right\}$$

$$Z\{f(n+k)\} = z^{k}\left[F(z) - f(0) - \frac{f(1)}{z} - \frac{f(2)}{z^{2}} - \cdots - \frac{f(k-1)}{z^{k-1}}\right]$$

Note: In the above result,

$$Z[f(n+1)] = z[F(z) - f(0)]$$

$$Z[f(n+2)] = z^{2}\left[F(z) - f(0) - \frac{f(1)}{z}\right]$$

$$Z[f(n+3)] = z^{3}\left[F(z) - f(0) - \frac{f(1)}{z} - \frac{f(2)}{z^{2}}\right]$$

These will be used in solving difference equations.

Change of Scale Property or Damping Rule

If $Z[f(n)] = F(z)$, then $Z[a^{-n}f(n)] = F(az) = \{Z[f(n)]\}_{z\to az} = \{F(z)\}_{z\to az}$

If $Z[f(n)] = F(z)$, then $Z[a^{n}f(n)] = F\left(\frac{z}{a}\right) = \{Z[f(n)]\}_{z\to\frac{z}{a}} = \{F(z)\}_{z\to\frac{z}{a}}$

Note:

In the differentiation, $d(u.v) = uv' + vu'$ and $d\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^{2}}$

Illustrative Examples:

Example 1: Find the Z-Transform of n^2

Solution: Given $f(n) = n^2$

We know that $Z[nf(n)] = -z\frac{d}{dz}[Z\{f(n)\}]$

$Z[n^2] = Z[n.n]$

$$= -z\frac{d}{dz}[Z(n)]$$

$$= -z\frac{d}{dz}\left[\frac{z}{(z-1)^{2}}\right] \because Z(n) = \left[\frac{z}{(z-1)^{2}}\right]$$

$$= -z\left[\frac{(z-1)^{2}.1 - z.2.(z-1)}{(z-1)^{4}}\right]$$

$$= -z\left[\frac{(z-1)\{z-1-2z\}}{(z-1)^{4}}\right]$$

$$= -z\left[\frac{(-z-1)}{(z-1)^{3}}\right]$$

$$Z[n^2] = \frac{z^{2}+z}{(z-1)^{3}}$$

Example 2: Find the Z-Transform of n^3

Solution: Given $f(n) = n^3$

We know that $Z[nf(n)] = -z\frac{d}{dz}[Z\{f(n)\}]$

$Z(n^3) = Z[n.n^2]$

$= -z\frac{d}{dz}[Z(n^2)]$ Since, $Z(n^2) = \frac{z^2+z}{(z-1)^3}$

$= -z\frac{d}{dz}\left[\frac{z^2+z}{(z-1)^3}\right]$

$= -z\left[\frac{(z-1)^3.(2z+1)-(z^2+z).3.(z-1)^2}{(z-1)^6}\right]$

$= -z\left[\frac{(z-1)^2[(z-1)(2z+1)-3.(z^2+z)]}{(z-1)^6}\right]$

$= -z\left[\frac{(z-1)(2z+1)-3.(z^2+z)}{(z-1)^4}\right]$

$= -z\left[\frac{2z^2-2z+z-1-3.z^2-3.z}{(z-1)^4}\right] = -z\left[\frac{-z^2-4z-1}{(z-1)^4}\right]$

$Z(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}$

Example 3: Find the Z-Transform of na^n

Solution: Given $f(n) = na^n$

We know that $Z[nf(n)] = -z\frac{d}{dz}[Z\{f(n)\}]$

$Z[na^n] = -z\frac{d}{dz}[Z\{a^n\}]$

$= -z\frac{d}{dz}\left[\frac{z}{z-a}\right] \because Z\{a^n\} = \frac{z}{z-a}$

$= -z\left[\frac{(z-a).1-z.1}{(z-a)^2}\right] = -z\left[\frac{z-a-z}{(z-a)^2}\right]$

$Z[na^n] = \frac{za}{(z-a)^2}$

Example 4: Find the Z-transform of $r^n \cos n\theta$ and $r^n \sin n\theta$

(Or)

Find $Z[r^n cos n\theta]$ and $Z[r^n sin n\theta]$

Solution:

We know that $Z[a^n] = \frac{z}{z-a}$

Taking $a = re^{i\theta}$

$a^n = r^n e^{in\theta} = r^n(cos n\theta + i \sin n\theta)$

$Z[r^n e^{in\theta}] = Z\left[\left(re^{i\theta}\right)^n\right]$

$Z[r^n\{cos n\theta + i \sin n\theta\}] = \frac{z}{z-re^{i\theta}}$

$Z[r^n cos n\theta + i\, r^n sin n\theta] = \frac{z}{z-r(cos\,\theta + i \sin\theta)}$

$Z[r^n cos n\theta] + iZ[r^n sin n\theta] = \frac{z}{z-rcos\,\theta - ir\, sin\,\theta}$

$= \frac{z}{(z-rcos\,\theta)-i\,rsin\,\theta} \times \frac{(z-rcos\,\theta)+i\,rsin\,\theta}{(z-rcos\,\theta)+i\,rsin\,\theta}$

$= \frac{z(z-rcos\,\theta)+i\,zrsin\,\theta}{(z-r\,cos\,\theta)^2 + r^2 sin^2\,\theta}$

$= \frac{z(z-rcos\,\theta)+i\,zrsin\,\theta}{z^2-2z\,r\,cos\,\theta + r^2 cos^2\,\theta + r^2 sin^2\,\theta}$

$= \frac{z(z-rcos\,\theta)+i\,zrsin\,\theta}{z^2-2zr\,cos\,\theta + r^2}$

$Z[r^n cos n\theta] + iZ[r^n sin n\theta] = \frac{z(z-rcos\,\theta)}{z^2-2zr\,cos\,\theta + r^2} + i\frac{zrsin\,\theta}{z^2-2zr\,cos\,\theta + r^2}$

Equating Real and Imaginary parts on both sides,

$Z[r^n cos n\theta] = \frac{z(z-rcos\,\theta)}{z^2-2z\,rcos\,\theta + r^2}$

$Z[r^n sin n\theta] = \frac{zrsin\,\theta}{z^2-2z\,rcos\,\theta + r^2}$

Example 5: Find the Z- transform of $cos n\theta$ and $sin\, n\theta$. Also find Z- transform of $cos\frac{n\pi}{2}$ and $sin\frac{n\pi}{2}$. (Or) Find $Z[cos n\theta]$ and $Z[sin n\theta]$. Also find $Z\left[cos\frac{n\pi}{2}\right]$ and $Z\left[sin\frac{n\pi}{2}\right]$

Solution: We know that $Z[a^n] = \frac{z}{z-a}$

$$\text{Taking } a = e^{i\theta}$$

$$a^n = e^{in\theta} = \cos n\theta + i \sin n\theta$$

$$Z[e^{in\theta}] = Z\left[\left(e^{i\theta}\right)^n\right]$$

$$Z[\cos n\theta + i \sin n\theta] = \frac{z}{z - e^{i\theta}}$$

$$Z[\cos n\theta] + iZ[\sin n\theta] = \frac{z}{z - (\cos\theta + i \sin\theta)}$$

$$= \frac{z}{z - \cos\theta - i \sin\theta}$$

$$= \frac{z}{(z - \cos\theta) - i \sin\theta} \times \frac{(z - \cos\theta) + i \sin\theta}{(z - \cos\theta) + i \sin\theta}$$

$$= \frac{z(z - \cos\theta) + i\, z \sin\theta}{(z - \cos\theta)^2 + \sin^2\theta}$$

$$= \frac{z(z - \cos\theta) + i\, z \sin\theta}{z^2 - 2z\cos\theta + \cos^2\theta + \sin^2\theta}$$

$$= \frac{z(z - \cos\theta) + i\, z \sin\theta}{z^2 - 2z\cos\theta + 1}$$

$$Z[\cos n\theta] + iZ[\sin n\theta] = \frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1} + i\frac{z \sin\theta}{z^2 - 2z\cos\theta + 1}$$

Equating Real and Imaginary parts on both sides,

$$Z[\cos n\theta] = \frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1} - - - - - - (1)$$

$$Z[\sin n\theta] = \frac{z \sin\theta}{z^2 - 2z\cos\theta + 1} - - - - - - (2)$$

Put $\theta = \frac{\pi}{2}$ in (1)&(2)we get

$$Z\left[\cos\frac{n\pi}{2}\right] = \frac{z\left(z - \cos\frac{\pi}{2}\right)}{z^2 - 2z\cos\frac{\pi}{2} + 1} = \frac{z(z-0)}{z^2 - 2z(0) + 1}$$

$$Z\left[\cos\frac{n\pi}{2}\right] = \frac{z^2}{z^2 + 1}$$

$$Z\left[\sin\frac{n\pi}{2}\right] = \frac{z\sin\frac{\pi}{2}}{z^2 - 2z\cos\frac{\pi}{2} + 1} = \frac{z(1)}{z^2 - 2z(0) + 1}$$

$$Z\left[\sin\frac{n\pi}{2}\right] = \frac{z}{z^2 + 1}$$

Example 6: Find $Z[2.3^n + 5(-2)^n]$

Solution:

$$Z[2.3^n + 5(-2)^n] = 2Z\{3^n\} + 5Z\{(-2)^n\}$$

$$= 2.\frac{z}{z-3} + 5\frac{z}{z+2}$$

Example 7: Find the Z-Transform of $(n+1)(n+2)$

Solution: Given $f(n) = (n+1)(n+2)$

$$= n^2 + 3n + 2$$

We know that $Z(n^2) = \frac{z^2 + z}{(z-1)^3}; Z(n) = \frac{z}{(z-1)^2}; Z(k) = \frac{k\,z}{z-1}$

$$Z[(n+1)(n+2)] = Z[n^2 + 3n + 2] = Z(n^2) + Z(3n) + Z(2)$$

$$= \frac{z^2 + z}{(z-1)^3} + \frac{3z}{(z-1)^2} + \frac{2z}{z-1}$$

$$= \frac{z^2 + z + 3z(z-1) + 2z(z-1)^2}{(z-1)^3}$$

$$= \frac{z^2 + z + 3z^2 - 3z + 2z(z^2 - 2z + 1)}{(z-1)^3}$$

$$= \frac{z^2 + z + 3z^2 - 3z + 2z^3 - 4z^2 + 2z}{(z-1)^3}$$

$$Z[(n+1)(n+2)] = \frac{2z^3}{(z-1)^3}$$

Example 8: Find $Z[n(n-1)(n-2)]$

Solution: Given $f(n) = n(n-1)(n-2) = n(n^2 - 3n + 2)$

$$= n^3 - 3n^2 + 2n$$

$$Z[f(n)] = Z[n^3 - 3n^2 + 2n]$$

$$= Z(n^3) - 3Z(n^2) + 2Z(n)$$

We know that, $Z(n^2) = \frac{z^2+z}{(z-1)^3}; Z(n) = \frac{z}{(z-1)^2}; Z(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}$

Now, $Z[n(n-1)(n-2)] = Z(n^3) - 3Z(n^2) + 2Z(n)$

$$= \frac{z^3+4z^2+z}{(z-1)^4} - 3\frac{z^2+z}{(z-1)^3} + 2\frac{z}{(z-1)^2}$$

$$= \frac{z^3+4z^2+z-3(z^2+z)(z-1)+2z(z-1)^2}{(z-1)^4}$$

$$= \frac{z^3+4z^2+z-3(z^3+z^2-z^2-z)+2z(z^2-2z+1)}{(z-1)^4}$$

$$= \frac{z^3+4z^2+z-3z^3-3z^2+3z^2+3z+2z^3-4z^2+2z}{(z-1)^4}$$

$$Z[n(n-1)(n-2)] = \frac{6z}{(z-1)^4}$$

Note:

$sin^3 A = \frac{3}{4}sin A - \frac{1}{4}sin 3A$	$cos^2 A = \frac{1+cos 2A}{2}$
$cos^3 A = \frac{3}{4}cos A + \frac{1}{4}cos 3A$	$sin^2 A = \frac{1-cos 2A}{2}$
$sin\frac{3\pi}{4} = \frac{1}{\sqrt{2}}; cos\frac{3\pi}{4} = \frac{-1}{\sqrt{2}}$	

Example 9: Find the Z-transform of $(i) sin^3\frac{n\pi}{4} (ii)\ cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right)$

Solution:

$$(i) sin^3\frac{n\pi}{4} = \frac{3}{4}\ sin\frac{n\pi}{4} - \frac{1}{4}sin\frac{3n\pi}{4}$$

$$Z\left\{sin^3\frac{n\pi}{4}\right\} = \frac{3}{4}Z\left\{\ sin\frac{n\pi}{4}\right\} - \frac{1}{4}Z\left\{sin\frac{3n\pi}{4}\right\}$$

$$= \frac{3}{4}\left\{\frac{zsin\frac{\pi}{4}}{z^2-2z\,cos\frac{\pi}{4}+1}\right\} - \frac{1}{4}\left\{\frac{zsin\frac{3\pi}{4}}{z^2-2z\,cos\frac{3\pi}{4}+1}\right\}$$

$$= \frac{3}{4}\left\{\frac{\frac{z}{\sqrt{2}}}{z^2-2z\left(\frac{1}{\sqrt{2}}\right)+1}\right\} - \frac{1}{4}\left\{\frac{\left(\frac{z}{\sqrt{2}}\right)}{z^2-2z\left(\frac{-1}{\sqrt{2}}\right)+1}\right\}$$

$$= \frac{3}{4\sqrt{2}}\left\{\frac{z}{z^2-\sqrt{2}z+1}\right\} - \frac{1}{4\sqrt{2}}\left\{\frac{z}{z^2+\sqrt{2}z+1}\right\}$$

(ii) We know that, $cos(A+B) = cosAcosB - sinAsinB$

$$cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right) = cos\frac{n\pi}{2}cos\frac{\pi}{4} - sin\frac{n\pi}{2}sin\frac{\pi}{4}$$

$$Z\left\{cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right)\right\} = Z\left\{cos\frac{n\pi}{2}\left(\frac{1}{\sqrt{2}}\right)\right\} - Z\left\{sin\frac{n\pi}{2}\left(\frac{1}{\sqrt{2}}\right)\right\}$$

$$= \frac{1}{\sqrt{2}}\left[\frac{z^2}{z^2+1} - \frac{z}{z^2+1}\right] = \frac{1}{\sqrt{2}}\left[\frac{z^2-z}{z^2+1}\right]$$

$$= \frac{1}{\sqrt{2}}\left[\frac{z(z-1)}{z^2+1}\right]$$

Example 10: Find the Z-Transform of $\frac{1}{n+1}$

Solution: Given $f(n) = \frac{1}{n+1}$

$$Z[f(n)] = \sum_{n=0}^{\infty} f(n)z^{-n}$$

$$Z\left[\frac{1}{n+1}\right] = \sum_{n=0}^{\infty}\frac{1}{n+1}z^{-n}$$

$$= \sum_{n=0}^{\infty}\frac{1}{n+1}\frac{1}{z^n} = \sum_{n=0}^{\infty}\frac{1}{n+1}\left(\frac{1}{z}\right)^n$$

$$= \sum_{n=0}^{\infty}\frac{\left(\frac{1}{z}\right)^n}{n+1}$$

$$= \frac{1}{1} + \frac{\left(\frac{1}{z}\right)}{2} + \frac{\left(\frac{1}{z}\right)^2}{3} + \frac{\left(\frac{1}{z}\right)^3}{4} + \cdots$$

$$= \frac{\left(\frac{1}{z}\right)}{\left(\frac{1}{z}\right)}\left[\frac{1}{1} + \frac{\left(\frac{1}{z}\right)}{2} + \frac{\left(\frac{1}{z}\right)}{3} + \frac{\left(\frac{1}{z}\right)}{4} + \cdots\right]$$

$$= \frac{1}{\left(\frac{1}{z}\right)}\left[\frac{\left(\frac{1}{z}\right)}{1} + \frac{\left(\frac{1}{z}\right)^2}{2} + \frac{\left(\frac{1}{z}\right)^3}{3} + \frac{\left(\frac{1}{z}\right)^4}{4} + \cdots\right]$$

$$= z\left[-\log\left(1-\frac{1}{z}\right)\right] = -z\log\left(\frac{z-1}{z}\right)$$

$$Z\left[\frac{1}{n+1}\right] = z\log\left(\frac{z}{z-1}\right)$$

Example 11: Find $Z\left[\frac{1}{(n+1)(n+2)}\right]$

Solution: Given $f(n) = \frac{1}{(n+1)(n+2)}$

Let $\frac{1}{(n+1)(n+2)} = \frac{A}{(n+1)} + \frac{B}{(n+2)} - - - -(1)$

Multiply by $(n+1)(n+2)$ on both sides

Let $\frac{(n+1)(n+2)}{(n+1)(n+2)} = \frac{A\,(n+1)(n+2)}{(n+1)} + \frac{B\,(n+1)(n+2)}{(n+2)}$

$1 = A(n+2) + B(n+1)$

$Put\ n = -1$ $1 = A(-1+2)$ $\Rightarrow A = 1$	$Put\ n = -2$ $1 = B(-2+1)$ $\Rightarrow B = -1$

(1) becomes, $\frac{1}{(n+1)(n+2)} = \frac{1}{(n+1)} - \frac{1}{(n+2)}$

$$Z\left[\frac{1}{(n+1)(n+2)}\right] = Z\left[\frac{1}{(n+1)} - \frac{1}{(n+2)}\right]$$

$$= Z\left[\frac{1}{(n+1)}\right] - Z\left[\frac{1}{(n+2)}\right] - - - - - -(2)$$

We know that $Z\left[\frac{1}{(n+1)}\right] = z\,log\left(\frac{z}{z-1}\right) - - - - - (3)$

$$Z\left[\frac{1}{(n+2)}\right] = \sum_{n=0}^{\infty}\frac{1}{n+2}z^{-n} = \sum_{n=0}^{\infty}\frac{1}{n+2}\frac{1}{z^n}$$

$$= \sum_{n=0}^{\infty}\frac{1}{n+2}\left(\frac{1}{z}\right)^n = \sum_{n=0}^{\infty}\frac{\left(\frac{1}{z}\right)^n}{n+2}$$

$$= \frac{1}{2} + \frac{\left(\frac{1}{z}\right)}{3} + \frac{\left(\frac{1}{z}\right)^2}{4} + \frac{\left(\frac{1}{z}\right)^3}{5} + \cdots$$

$$= \frac{\left(\frac{1}{z}\right)^2}{\left(\frac{1}{z}\right)^2}\left[\frac{1}{2} + \frac{\left(\frac{1}{z}\right)}{3} + \frac{\left(\frac{1}{z}\right)^2}{4} + \frac{\left(\frac{1}{z}\right)^3}{5} + \cdots\right]$$

$$= \frac{1}{\left(\frac{1}{z}\right)^2}\left[\left\{\frac{\left(\frac{1}{z}\right)}{1} + \frac{\left(\frac{1}{z}\right)^2}{2} + \frac{\left(\frac{1}{z}\right)^3}{3} + \frac{\left(\frac{1}{z}\right)^4}{4} + \cdots\right\} - \frac{\left(\frac{1}{z}\right)}{1}\right]$$

$$= \frac{1}{\left(\frac{1}{z^2}\right)}\left[\left\{-\log\left(1-\frac{1}{z}\right)\right\} - \frac{1}{z}\right]$$

$$= z^2\left[-\log\left(\frac{z-1}{z}\right) - \frac{1}{z}\right] = z^2\left[\log\left(\frac{z}{z-1}\right) - \frac{1}{z}\right]$$

$$Z\left[\frac{1}{(n+2)}\right] = z^2\log\left(\frac{z}{z-1}\right) - z \ldots\ldots\ldots\ldots (4)$$

Sub. (3) & (4) in (2),

$$Z\left[\frac{1}{(n+1)(n+2)}\right] = z\,log\left(\frac{z}{z-1}\right) - \left[z^2 log\left(\frac{z}{z-1}\right) - z\right]$$

$$= z\,log\left(\frac{z}{z-1}\right) - z^2 log\left(\frac{z}{z-1}\right) + z$$

$$Z\left[\frac{1}{(n+1)(n+2)}\right] = (z - z^2)\,log\left(\frac{z}{z-1}\right) + z$$

Example 12: Find $Z\left[\frac{2n+3}{(n+1)(n+2)}\right]$

Solution: Given $f(n) = \frac{}{(n+1)(n+2)}$

Let $\frac{2n+3}{(n+1)(n+2)} = \frac{A}{(n+1)} + \frac{B}{(n+2)} ----(1)$

Multiply by $(n+1)(n+2)$ on both sides,

$$\frac{2n+3}{(n+1)(n+2)}(n+1)(n+2) = \frac{A(n+1)(n+2)}{(n+1)} + \frac{B(n+1)(n+2)}{(n+2)}$$

$$2n+3 = A(n+2) + B(n+1)$$

$Put\ n = -1$ $-2+3 = A(-1+2) + B(-1+1)$ $\Rightarrow A = 1$	$Put\ n = -2$ $-4+3 = A(-2+2) + B(-2+1)$ $\Rightarrow B = 1$

(1) becomes, $\frac{2n+3}{(n+1)(n+2)} = \frac{1}{(n+1)} + \frac{1}{(n+2)}$

$$Z\left[\frac{2n+3}{(n+1)(n+2)}\right] = Z\left[\frac{1}{(n+1)} + \frac{1}{(n+2)}\right]$$

$$= Z\left[\frac{1}{(n+1)}\right] + Z\left[\frac{1}{(n+2)}\right] \ldots\ldots\ldots\ldots (2)$$

We know that, $Z\left[\frac{1}{(n+1)}\right] = z\,log\left(\frac{z}{z-1}\right) \ldots\ldots\ldots\ldots (3)$

$$Z\left[\frac{1}{(n+2)}\right] = \sum_{n=0}^{\infty} \frac{1}{n+2} z^{-n} = \sum_{n=0}^{\infty} \frac{1}{n+2} \frac{1}{z^n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n+2} \left(\frac{1}{z}\right)^n = \sum_{n=0}^{\infty} \frac{\left(\frac{1}{z}\right)^n}{n+2}$$

$$= \frac{1}{2} + \frac{\left(\frac{1}{z}\right)}{3} + \frac{\left(\frac{1}{z}\right)^2}{4} + \frac{\left(\frac{1}{z}\right)^3}{5} + \cdots$$

$$= \frac{\left(\frac{1}{z}\right)^2}{\left(\frac{1}{z}\right)^2}\left[\frac{1}{2} + \frac{\left(\frac{1}{z}\right)}{3} + \frac{\left(\frac{1}{z}\right)^2}{4} + \frac{\left(\frac{1}{z}\right)^3}{5} + \cdots\right]$$

$$= \frac{1}{\left(\frac{1}{z}\right)^2}\left[\left\{\frac{\left(\frac{1}{z}\right)}{1} + \frac{\left(\frac{1}{z}\right)^2}{2} + \frac{\left(\frac{1}{z}\right)^3}{3} + \frac{\left(\frac{1}{z}\right)^4}{4} + \cdots\right\} - \frac{\left(\frac{1}{z}\right)}{1}\right]$$

$$= \frac{1}{\left(\frac{1}{z^2}\right)}\left[\left\{-log\left(1-\frac{1}{z}\right)\right\} - \frac{1}{z}\right]$$

$$= z^2\left[-log\left(\frac{z-1}{z}\right) - \frac{1}{z}\right] = z^2\left[log\left(\frac{z}{z-1}\right) - \frac{1}{z}\right]$$

$$Z\left[\frac{1}{(n+2)}\right] = z^2 log\left(\frac{z}{z-1}\right) - z \ldots\ldots\ldots\ldots (4)$$

Sub. (3) & (4) in (2),

$$Z\left[\frac{2n+3}{(n+1)(n+2)}\right] = z\,log\left(\frac{z}{z-1}\right) + \left[z^2 log\left(\frac{z}{z-1}\right) - z\right]$$

$$= (z+z^2)\,log\left(\frac{z}{z-1}\right) - z$$

Problems for Practice:

1. Find $Z\left\{\frac{(n+1)(n+2)}{2}\right\}$ Answer: $\frac{z^3}{(z-1)^3}$
2. Find Z-transform of $(n+1)^2$ Answer: $\frac{z^2(z+1)}{(z-1)^3}$
3. Find $Z\left\{\frac{1}{n(n-1)}\right\}$ Answer: $\left(\frac{z-1}{z}\right) log\left(\frac{z-1}{z}\right)$

Unit Impulse Function (Or) Unit Sample Sequence

The unit impulse function $\delta(n)$ is defined as the sequence with values,

$$\delta(n) = \begin{cases} 1 & if\ n = 0 \\ 0 & if\ n \neq 0 \end{cases}$$

Z-transform of Unit Impulse Function

$$Z\{\delta(n)\} = \sum_{n=0}^{\infty} \delta(n) z^{-n} = 1 + 0 + 0 + \cdots$$

$$Z\{\delta(n)\} = 1$$

Unit Step Sequence

The unit step sequence $u(n)$ is defined as the sequence with values,

$$u(n) = \begin{cases} 1 \; if \; n = 0,1,2,\ldots \\ 0 \; if \; n < 0 \end{cases}$$

Z-transform of Unit Step Sequence

$$Z\{u(n)\} = \sum_{n=0}^{\infty} u(n) z^{-n} = \sum_{n=0}^{\infty} (1) \frac{1}{z^n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^n = 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \cdots$$

$$= \left(1 - \frac{1}{z}\right)^{-1} = \left(\frac{z-1}{z}\right)^{-1}$$

$$Z\{u(n)\} = \frac{z}{z-1}$$

Illustrative Example: Find $Z[n(n-1)u(n)]$

Solution: The unit step sequence $u(n)$ is defined as,

$$u(n) = \begin{cases} 1 \; if \; n = 0,1,2,\ldots \\ 0 \; if \; n < 0 \end{cases}$$

Given $f(n) = n(n-1)u(n) = n(n-1).1$

$= n(n-1) \; if \; n = 0,1,2\ldots$

$Z[f(n)] = Z[n(n-1)] = Z[n^2 - n] = Z(n^2) - Z(n)$

We know that, $Z(n^2) = \frac{z^2 + z}{(z-1)^3}$; $Z(n) = \frac{z}{(z-1)^2}$

$$Z[f(n)] = Z(n^2) - Z(n)$$

$$= \frac{z^2 + z}{(z-1)^3} - \frac{z}{(z-1)^2}$$

$$= \frac{(z^2 + z) - z(z-1)}{(z-1)^3} = \frac{z^2 + z - z^2 + z}{(z-1)^3}$$

$$Z[f(n)] = \frac{2z}{(z-1)^3}$$

Initial and Final Value Theorem

Initial Value Theorem:

If $Z[f(n)] = F(z)$ then $f(0) = \lim_{z \to \infty} F(z)$

Proof:

Given $F(z) = Z[f(n)]$

$$= \sum_{n=0}^{\infty} f(n) z^{-n} = \sum_{n=0}^{\infty} f(n) \frac{1}{z^n}$$

$$= \sum_{n=0}^{\infty} \frac{f(n)}{z^n}$$

$$F(z) = f(0) + \frac{f(1)}{z} + \frac{f(2)}{z^2} + \frac{f(3)}{z^3} + \cdots$$

Taking $\lim_{z \to \infty}$ on both sides,

$$\lim_{z \to \infty} F(z) = \lim_{z \to \infty} \left[f(0) + \frac{f(1)}{z} + \frac{f(2)}{z^2} + \frac{f(3)}{z^3} + \cdots \right]$$

$$\lim_{z \to \infty} F(z) = f(0) + 0 + 0 + \cdots \qquad \left\{ \text{Since } \lim_{z \to \infty} \frac{1}{z} = 0 \right\}$$

Hence, $f(0) = \lim_{z \to \infty} F(z)$

Note:

$$f(1) = \lim_{z \to \infty} z\{F(z) - f(0)\}$$

$$f(2) = \lim_{z \to \infty} z^2\{F(z) - f(0) - f(1)z^{-1}\}$$

$$f(3) = \lim_{z \to \infty} z^3\{F(z) - f(0) - f(1)z^{-1} - f(2)z^{-2}\}$$

Final Value Theorem:

If $Z[f(n)] = F(z)$ then $\lim_{n \to \infty} f(n) = \lim_{z \to 1} [(z-1)F(z)]$

Proof:

We know that $Z[f(n+1)] = z[F(z) - f(0)]$

$$Z[f(n+1)] = zF(z) - zf(0)$$

$$Z[f(n+1)] - F(z) = zF(z) - F(z) - zf(0)$$

$$Z[f(n+1)] - Z[f(n)] = (z-1)F(z) - zf(0)$$

$$Z[f(n+1) - f(n)] = (z-1)F(z) - zf(0)$$

By definition, $Z[f(n)] = F(z) = \sum_{n=0}^{\infty} f(n)z^{-n}$

$$\sum_{n=0}^{\infty}\{f(n+1) - f(n)\}z^{-n} = (z-1)F(z) - zf(0)$$

$$(z-1)F(z) - zf(0) = \sum_{n=0}^{\infty}\{f(n+1) - f(n)\}z^{-n}$$

Taking $\lim_{z\to 1}$ on both sides,

$$\lim_{z\to 1}\{(z-1)F(z) - zf(0)\} = \lim_{z\to 1}\sum_{n=0}^{\infty}\{f(n+1) - f(n)\}z^{-n}$$

$$\lim_{z\to 1}(z-1)F(z) - f(0) = \sum_{n=0}^{\infty}\{f(n+1) - f(n)\}$$

$$= \lim_{n\to\infty}[\{f(n+1) - f(n)\} + \{f(n) - f(n-1)\}$$

$$+ \{f(n-1) - f(n-2)\} + \cdots + \{f(1) - f(0)\}]$$

$$= \lim_{n\to\infty}\{f(n+1) - f(0)\}$$

$$= f(\infty) - f(0)$$

$$\lim_{z\to 1}(z-1)F(z) - f(0) = \lim_{n\to\infty} f(n) - f(0)$$

$$\lim_{n\to\infty} f(n) = \lim_{z\to 1}(z-1)F(z)$$

Illustrative Examples:

Example 1: If $F(z) = Z\{f(n)\}$ find $f(0)$ and $f(1)$, where $F(z) = \frac{z}{z-1} + \frac{z}{z^2+1}$

Solution: Given $F(z) = \frac{z}{z-1} + \frac{z}{z^2+1}$

By Initial Value Theorem, $f(0) = \lim_{z\to\infty} F(z)$

$$= \lim_{z\to\infty}\left\{\frac{z}{z-1} + \frac{z}{z^2+1}\right\} = \lim_{z\to\infty}\left\{\frac{z}{z\left(1-\frac{1}{z}\right)} + \frac{z}{z\left(z+\frac{1}{z}\right)}\right\}$$

$$= \lim_{z\to\infty}\left\{\frac{1}{1-\frac{1}{z}} + \frac{1}{z+\frac{1}{z}}\right\} = 1 + 0$$

$$f(0) = 1$$

We know that, $f(1) = \lim_{z\to\infty} z\left\{\frac{z}{z-1} + \frac{z}{z^2+1} - f(0)\right\}$

$$= \lim_{z\to\infty} z\left\{\frac{z}{z-1} + \frac{z}{z^2+1} - 1\right\} = \lim_{z\to\infty} z\left\{\frac{z}{z-1} - 1 + \frac{z}{z^2+1}\right\}$$

$$= \lim_{z\to\infty} z\left\{\frac{z-z+1}{z-1} + \frac{z}{z^2+1}\right\} = \lim_{z\to\infty} z\left\{\frac{1}{z-1} + \frac{z}{z^2+1}\right\}$$

$$= \lim_{z\to\infty}\left\{\frac{z}{z-1} + \frac{z^2}{z^2+1}\right\} = \lim_{z\to\infty}\left\{\frac{z}{z\left(1-\frac{1}{z}\right)} + \frac{z^2}{z^2\left(1+\frac{1}{z^2}\right)}\right\}$$

$$= \lim_{z\to\infty}\left\{\frac{1}{1-\frac{1}{z}} + \frac{1}{1+\frac{1}{z^2}}\right\} = 1 + 1$$

$$f(1) = 2$$

Example 2: Verify initial value theorem for $f(n) = \frac{2^{n+1}}{n!}$

Solution:

By Initial Value Theorem, $f(0) = \lim_{z\to\infty} F(z)$

Given $f(n) = \frac{2^{n+1}}{n!}$

$$f(0) = 2 - - - -(1)$$

$$F(z) = Z\{f(n)\}$$

$$= Z\left\{\frac{2^{n+1}}{n!}\right\} = Z\left\{\frac{2.2^n}{n!}\right\}$$

$$= 2\,Z\left\{\frac{2^n}{n!}\right\} = 2e^{\frac{2}{z}} \qquad \left\{\text{By result, } Z\left\{\frac{a^n}{n!}\right\} = e^{\frac{a}{z}}\right\}$$

$$\lim_{z\to\infty} F(z) = \lim_{z\to\infty} 2e^{\frac{2}{z}} = 2e^0$$

$$\lim_{z\to\infty} F(z) = 2 ----(2)$$

From (1)&(2),

$$f(0) = \lim_{z\to\infty} F(z) = 2$$

Hence, initial value theorem is verified.

Example 3: Verify final value theorem for $f(n) = 2^{-n}\sin\frac{n\pi}{2}$

Solution: By Final Value Theorem, $\lim_{n\to\infty} f(n) = \lim_{z\to 1}[(z-1)F(z)]$

Given $f(n) = 2^{-n}\sin\frac{n\pi}{2}$

$$\lim_{n\to\infty} f(n) = \lim_{n\to\infty} 2^{-n}\sin\frac{n\pi}{2}$$

$$= \lim_{n\to\infty} \frac{1}{2^n}\sin\frac{n\pi}{2}$$

$$\lim_{n\to\infty} f(n) = 0 ------(1)$$

$$F(z) = Z\{f(n)\}$$

$$= Z\left\{2^{-n}\sin\frac{n\pi}{2}\right\}$$

$$= Z\left\{\left(\frac{1}{2}\right)^n \sin\frac{n\pi}{2}\right\}$$

$$= \frac{\frac{z}{2}}{z^2+\frac{1}{4}} = \frac{\frac{z}{2}}{\frac{4z^2+1}{4}}$$

$$F(z) = \frac{2z}{4z^2+1}$$

Put $\theta = \frac{\pi}{2}$ and $r = \frac{1}{2}$ in the result

$Z[r^n \sin n\theta] = \frac{zr\sin\theta}{z^2 - 2zr\cos\theta + r^2}$ weget,

$$Z\left[\left(\frac{1}{2}\right)^n \sin\frac{n\pi}{2}\right] = \frac{z\left(\frac{1}{2}\right)\sin\frac{\pi}{2}}{z^2 - 2z\left(\frac{1}{2}\right)\cos\frac{\pi}{2} + \left(\frac{1}{2}\right)^2} = \frac{\frac{z}{2}}{z^2+\frac{1}{4}}$$

$$\lim_{z\to 1}[(z-1)F(z)] = \lim_{z\to 1}(z-1)\frac{2z}{4z^2+1}$$

$$\lim_{z\to 1}[(z-1)F(z)] = 0 ----(2)$$

From (1)&(2), $\lim_{n\to\infty} f(n) = \lim_{z\to 1}[(z-1)F(z)] = 0$

Hence, final value theorem is verified.

INVERSE Z- TRANSFORMS

Definition

If $Z[f(n)] = F(z)$, then $F(z)$ is unique and the inverse Z-transform of $F(z)$ is $Z^{-1}\{F(z)\} = f(n)$.

Table of Inverse Z-Transforms

Sl. No.	$f(n)$	$Z\{f(n)\} = F(z)$	$Z^{-1}[F(z)]$	$f(n)$
1.	a^n	$\frac{z}{z-a}$	$Z^{-1}\left[\frac{z}{z-a}\right]$	a^n
2.	a^{n-1}	$\frac{1}{z-a}$	$Z^{-1}\left[\frac{z}{z-a}\right]$	a^{n-1} ,$n \geq 1$
3.	n	$\frac{z}{(z-1)^2}$	$Z^{-1}\left[\frac{z}{(z-1)^2}\right]$	n
4.	na^n	$\frac{az}{(z-a)^2}$	$Z^{-1}\left[\frac{az}{(z-a)^2}\right]$	az
5.	na^{n-1}	$\frac{z}{(z-a)^2}$	$Z^{-1}\left[\frac{z}{(z-a)^2}\right]$	na^{n-1}
6.	$a^n\cos\frac{n\pi}{2}$	$\frac{z^2}{z^2+a^2}$	$Z^{-1}\left[\frac{z^2}{z^2+a^2}\right]$	$a^n\cos\frac{n\pi}{2}$
7.	$a^n\sin\frac{n\pi}{2}$	$\frac{az}{z^2+a^2}$	$Z^{-1}\left[\frac{az}{z^2+a^2}\right]$	$a^n\sin\frac{n\pi}{2}$

Convolution of Two Sequences

The convolution of two sequences $\{f(n)\}$ and $\{g(n)\}$ is defined as

$$f(n) * g(n) = \sum_{k=0}^{n} f(k)\,g(n-k)$$

Note: $f(n) * g(n) = g(n) * f(n)$

Convolution Theorem:

If $F(z)$ and $G(z)$ are the Z-transforms of $f(n)$ and $g(n)$ respectively, then

$Z\{f(n) * g(n)\} = F(Z)G(Z)$. Where $f(n) * g(n)$ is the convolution *of* $f(n)$ and $g(n)$.

Also, $Z^{-1}\{F(Z)G(Z)\} = f(n) * g(n)$

Proof: By definition of Z-transforms,

$$Z[f(n)] = F(z) = \sum_{n=0}^{\infty} f(n)z^{-n}$$

$$\text{and } Z[g(n)] = G(z) = \sum_{n=0}^{\infty} g(n)z^{-n}$$

We have $F(z)G(z) = \left[\sum_{n=0}^{\infty} f(n)\, z^{-n}\right]\left[\sum_{n=0}^{\infty} g(n)z^{-n}\right]$

$$= [f(0) + f(1)z^{-1} + f(2)z^{-2} + \cdots + f(n)z^{-n} + \cdots][g(0) + g(1)z^{-1} + g(2)z^{-2} + \cdots + g(n)z^{-n} + \cdots]$$

Multiplying and Collecting the co-efficient of z^{-n} we get,

$$F(z)G(z) = \sum_{n=0}^{\infty}[f(0)g(n) + f(1)g(n-1) + f(2)g(n-2) + \cdots + f(n)g(0)]\, z^{-n}$$

By definition,

$$F(z)G(z) = Z[f(0)g(n) + f(1)g(n-1) + f(2)g(n-2) + \cdots + f(n)g(0)]$$

$$F(z)G(z) = \sum_{k=0}^{n}\{f(k)g(n-k)\}z^{-n}$$

$$F(z)G(z) = Z\{f(n) * g(n)\}$$

Hence, $f(n) * g(n) = Z^{-1}\{F(z)G(z)\}$

Note: $Z^{-1}[F(z)G(z)] = Z^{-1}[f(z)] * Z^{-1}[g(z)]$

Illustrative Examples:

Example 1: Using convolution theorem, evaluate $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right]$.

Solution:

We know that $Z(a^n) = \frac{z}{z-a} \Rightarrow Z^{-1}\left[\frac{z}{z-a}\right] = a^n$

$$Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right] = Z^{-1}\left[\frac{z}{z-a}\frac{z}{z-b}\right]$$

$$= Z^{-1}\left[\frac{z}{z-a}\right] * Z^{-1}\left[\frac{z}{z-b}\right]$$

$$= a^n * b^n$$

$$= \sum_{k=0}^{n} a^k.b^{n-k} \left[\because f(n) * g(n) = \sum_{k=0}^{n} f(k)g(n-k)\right]$$

$$= \sum_{k=0}^{n} a^k b^n b^{-k} = b^n \sum_{k=0}^{n} a^k b^{-k}$$

$$= b^n \sum_{k=0}^{n} a^k \left(\frac{1}{b^k}\right) = b^n \sum_{k=0}^{n} \left(\frac{a}{b}\right)^k$$

$$= b^n\left[1 + \left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2 + \cdots + \left(\frac{a}{b}\right)^n\right]$$

$$= b^n\left[\frac{\left(\frac{a}{b}\right)^{n+1} - 1}{\frac{a}{b} - 1}\right] \left[\because \text{Sum of G.P.} = \left(\frac{r^{n+1}-1}{r-1}\right), \text{where } r = \frac{a}{b}\right.$$

$$= b^n\left[\frac{\frac{a^{n+1}}{b^{n+1}} - 1}{\frac{a}{b} - 1}\right] = b^n\left[\frac{\left(\frac{a^{n+1} - b^{n+1}}{b^{n+1}}\right)}{\left(\frac{a-b}{b}\right)}\right]$$

$$= b^n\left[\frac{a^{n+1} - b^{n+1}}{b^{n+1}}\frac{b}{a-b}\right]$$

$$= b^{n+1}\left[\frac{a^{n+1} - b^{n+1}}{b^{n+1}(a-b)}\right] = \frac{b^{n+1}}{b^{n+1}}\left[\frac{a^{n+1} - b^{n+1}}{a-b}\right]$$

Hence, $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right] = \frac{a^{n+1} - b^{n+1}}{a-b}$

Example 2: Using convolution theorem, find $Z^{-1}\left[\frac{z^2}{(z-1)(z-3)}\right]$

Solution:

We know that, $Z(a^n)=\frac{z}{z-a}$

$$Z^{-1}\left[\frac{z^2}{(z-1)(z-3)}\right]=Z^{-1}\left[\frac{z}{z-1}\frac{z}{z-3}\right]$$

$$=Z^{-1}\left[\frac{z}{z-1}\right]*Z^{-1}\left[\frac{z}{z-3}\right]$$

$$=1^n*3^n$$

$$=\sum_{k=0}^{n}1^k3^{n-k}=\sum_{k=0}^{n}3^{n-k}$$

$$=3^n+3^{n-1}+\cdots+3+3^0$$

$$=1+3+3^2+\cdots+3^n$$

$$=\frac{3^{n+1}-1}{3-1}\qquad \text{For Sum of G.P.}=\left(\frac{r^{n+1}-1}{r-1}\right),r>1$$

$$=\frac{1}{2}(3^{n+1}-1)$$

$$Z^{-1}\left[\frac{z^2}{(z-1)(z-3)}\right]=\frac{1}{2}[3^{n+1}-1]$$

Example 3: Find $Z^{-1}\left[\frac{8z^2}{(2z-1)(4z-1)}\right]$, using convolution theorem

Solution:

$$Z^{-1}\left[\frac{8z^2}{(2z-1)(4z-1)}\right]=Z^{-1}\left[\frac{8z^2}{2\left(z-\frac{1}{2}\right).4\left(z-\frac{1}{4}\right)}\right]$$

$$=Z^{-1}\left[\frac{8z^2}{8\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\right]=Z^{-1}\left[\frac{z^2}{\left(z-\frac{1}{2}\right)\left(z-\frac{1}{4}\right)}\right]$$

$$=Z^{-1}\left[\frac{z}{\left(z-\frac{1}{2}\right)}.\frac{z}{\left(z-\frac{1}{4}\right)}\right]$$

$$=Z^{-1}\left[\frac{z}{\left(z-\frac{1}{2}\right)}\right]*Z^{-1}\left[\frac{z}{\left(z-\frac{1}{4}\right)}\right]$$

$$=\left(\frac{1}{2}\right)^n*\left(\frac{1}{4}\right)^n$$

$$=\sum_{k=0}^{n}\left(\frac{1}{4}\right)^k\left(\frac{1}{2}\right)^{n-k}$$

$$=\sum_{k=0}^{n}\left(\frac{1}{4}\right)^k\left(\frac{1}{2}\right)^n\left(\frac{1}{2}\right)^{-k}$$

$$=\left(\frac{1}{2}\right)^n\sum_{k=0}^{n}\left(\frac{1}{4}\right)^k\left(\frac{1}{2}\right)^{-k}$$

$$=\left(\frac{1}{2}\right)^n\sum_{k=0}^{n}\left(\frac{1}{4}\right)^k2^k\left[\because\left(\frac{1}{2}\right)^{-k}=2^k\right]$$

$$=\left(\frac{1}{2}\right)^n\sum_{k=0}^{n}\left(\frac{1}{4}.2\right)^k=\left(\frac{1}{2}\right)^n\sum_{k=0}^{n}\left(\frac{1}{2}\right)^k$$

$$=\left(\frac{1}{2}\right)^n\left\{1+\left(\frac{1}{2}\right)+\left(\frac{1}{2}\right)^2+\cdots+\left(\frac{1}{2}\right)^n\right\}$$

$$=\left(\frac{1}{2}\right)^n\left[\frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}}\right]\text{For Sum of G.P.}=\left(\frac{1-r^{n+1}}{1-r}\right),r<1$$

$$=\left(\frac{1}{2}\right)^n\left[\frac{1-\left(\frac{1}{2}\right)^{n+1}}{\frac{1}{2}}\right]=\left(\frac{1}{2}\right)^{n-1}\left[1-\left(\frac{1}{2}\right)^{n+1}\right]$$

$$Z^{-1}\left[\frac{8z^2}{(2z-1)(4z-1)}\right]=\left(\frac{1}{2}\right)^{n-1}-\left(\frac{1}{2}\right)^{2n}$$

Example 4: Using convolution theorem to find the inverse Z-transform $\frac{12}{(3-z^{-1})(4-z^{-1})}$

Solution:

$$Z^{-1}\left[\frac{12}{(3-z^{-1})(4-z^{-1})}\right] = Z^{-1}\left[\frac{12}{\left(3-\frac{1}{z}\right)\left(4-\frac{1}{z}\right)}\right]$$

$$= Z^{-1}\left[\frac{12}{\left(\frac{3z-1}{z}\right)\left(\frac{4z-1}{z}\right)}\right] = Z^{-1}\left[\frac{12}{\frac{(3z-1)(4z-1)}{z^2}}\right]$$

$$= Z^{-1}\left[\frac{12z^2}{3\left(z-\frac{1}{3}\right)4\left(z-\frac{1}{4}\right)}\right]$$

$$= Z^{-1}\left[\frac{z^2}{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{4}\right)}\right]$$

$$= Z^{-1}\left[\frac{z}{z-\frac{1}{3}}\right] * Z^{-1}\left[\frac{z}{z-\frac{1}{4}}\right]$$

$$= \left(\frac{1}{3}\right)^n * \left(\frac{1}{4}\right)^n$$

$$= \sum_{k=0}^{n}\left(\frac{1}{3}\right)^k\left(\frac{1}{4}\right)^{n-k}$$

$$= \sum_{k=0}^{n}\left(\frac{1}{3}\right)^k\left(\frac{1}{4}\right)^{-k}\left(\frac{1}{4}\right)^n$$

$$= \sum_{k=0}^{n}\left(\frac{1}{3}\right)^k(4)^k\left(\frac{1}{4}\right)^n = \left(\frac{1}{4}\right)^n\sum_{k=0}^{n}\left(\frac{4}{3}\right)^k$$

$$= \left(\frac{1}{4}\right)^n\left[1+\left(\frac{4}{3}\right)+\left(\frac{4}{3}\right)^2+\cdots+\left(\frac{4}{3}\right)^n\right]$$

$$= \left(\frac{1}{4}\right)^n\left[\frac{\left(\frac{4}{3}\right)^{n+1}-1}{\frac{4}{3}-1}\right] = \left(\frac{1}{4}\right)^n\left[\frac{\frac{4^{n+1}}{3^{n+1}}-1}{\frac{4}{3}-1}\right]$$

$$= \left(\frac{1}{4}\right)^n\frac{\left[\frac{4^{n+1}-3^{n+1}}{3^{n+1}}\right]}{\left[\frac{4-3}{3}\right]}$$

$$= \left(\frac{1}{4}\right)^n\frac{\left(\frac{4^{n+1}-3^{n+1}}{3^{n+1}}\right)}{\left(\frac{1}{3}\right)} = \frac{1}{4^n}\left(\frac{4^{n+1}-3^{n+1}}{3^n.3}\right)3$$

$$= \frac{1}{4^n}\left(\frac{4^{n+1}-3^{n+1}}{3^n}\right)$$

$$= \frac{1}{12^n}[4^n.4-3^n.3] = \frac{4^n.4}{12^n}-\frac{3^n.3}{12^n}$$

$$Z^{-1}\left[\frac{12}{(3-z^{-1})(4-z^{-1})}\right] = 4\left(\frac{1}{3}\right)^n - 3\left(\frac{1}{4}\right)^n$$

Example 5: Find $Z^{-1}\left[\frac{z^2}{(z+a)^2}\right]$, using convolution theorem.

Solution:

$$Z^{-1}\left[\frac{z^2}{(z+a)^2}\right] = Z^{-1}\left[\frac{z}{z+a}.\frac{z}{z+a}\right]$$

$$= Z^{-1}\left[\frac{z}{z+a}\right] * Z^{-1}\left[\frac{z}{z+a}\right]$$

$$= (-a)^n * (-a)^n$$

$$= \sum_{k=0}^{n}(-a)^k(-a)^{n-k} = \sum_{k=0}^{n}(-a)^k(-a)^n(-a)^{-k}$$

$$= (-a)^n\sum_{k=0}^{n}(-a)^k(-a)^{-k}$$

$$= (-a)^n\sum_{k=0}^{n}(-a)^{k-k} = (-a)^n\sum_{k=0}^{n}(-a)^0$$

$$= (-a)^n\sum_{k=0}^{n}(1) = (-a)^n\sum_{k=0}^{n}(1)$$

$$= (-a)^n\{1+1+\cdots+1\ \ (n+1)times\}$$

$$= (-a)^n (n+1)$$

$$Z^{-1}\left[\frac{z^2}{(z+a)^2}\right] = (n+1)(-a)^n$$

Example 6: Find $Z^{-1}\left[\left(\frac{z}{z-1}\right)^3\right]$, using convolution theorem.

Solution:

$$Z^{-1}\left[\left(\frac{z}{z-1}\right)^3\right] = Z^{-1}\left[\left(\frac{z}{z-1}\right)\left(\frac{z}{z-1}\right)^2\right]$$

$$= Z^{-1}\left[\left(\frac{z}{z-1}\right)\frac{z^2}{(z-1)^2}\right]$$

$$= Z^{-1}\left[\frac{z}{z-1}\right] * Z^{-1}\left[\frac{z^2}{(z-1)^2}\right] \quad \text{Since, } Z^{-1}\left[\frac{z^2}{(z-1)^2}\right] = (n+1)(1)^n$$

$$= (1)^n * (n+1)(1)^n$$

$$= (n+1)(1)^n * (1)^n$$

$$= \sum_{k=0}^{n} (k+1)(1)^k (1)^{n-k}$$

$$= \sum_{k=0}^{n} (k+1)$$

$$= 1 + 2 + 3 + \cdots + (n+1) \qquad \text{Since, } 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

$$Z^{-1}\left[\left(\frac{z}{z-1}\right)^3\right] = \frac{(n+1)(n+2)}{2}$$

Problems for Practice

1. Using convolution theorem, find $Z^{-1}\left[\frac{8z^2}{(2z-1)(4z+1)}\right]$ Ans: $\frac{2}{3}\left(\frac{1}{2}\right)^n + \frac{1}{3}\left(-\frac{1}{4}\right)^n$

2. Using convolution theorem, find the inverse Z-transform of $\left(\frac{z}{z-a}\right)^2$ Ans: $(n+1)a^n$

Inverse Z-Transform by Method of Partial Fraction

When $F(z)$ *is* a rational function in which the denominator can be factorised, $F(z)$ is resolved into partial fractions and then $Z^{-1}\{F(z)\}$ is derived as the sum of the inverse Z-transforms of the partial fractions.

Illustrative Examples:

Example 1: Find the inverse Z-transform of $\frac{2z^2+3z}{(z+2)(z-4)}$

Solution: Let $F(z) = \frac{2z^2 + 3z}{(z+2)(z-4)}$

$$F(z) = \frac{z(2z+3)}{(z+2)(z-4)}$$

$$\frac{F(z)}{z} = \frac{2z+3}{(z+2)(z-4)} - - - - - - - (1)$$

$$\frac{2z+3}{(z+2)(z-4)} = \frac{A}{z+2} + \frac{B}{z-4} - - - - - - (2)$$

Multiply by $(z+2)(z-4)$

$$\frac{2z+3}{(z+2)(z-4)}(z+2)(z-4) = \frac{A(z+2)(z-4)}{z+2} + \frac{B(z+2)(z-4)}{z-4}$$

$$2z + 3 = A(z-4) + B(z+2)$$

$Put\ z = 4$	$Put\ z = -2$
$2(4) + 3 = B(4+2)$	$-4 + 3 = A(-2-4)$
$6B = 11 \Rightarrow B = \frac{11}{6}$	$-6A = -1 \Rightarrow A = \frac{1}{6}$

(2)becomes, $\frac{2z+3}{(z+2)(z-4)} = \frac{\frac{1}{6}}{z+2} + \frac{\frac{11}{6}}{z-4}$

$$\frac{F(z)}{z} = \frac{1}{6(z+2)} + \frac{11}{6(z-4)} \qquad \text{Using (1)}$$

$$F(z) = \frac{z}{6(z+2)} + \frac{11\,z}{6(z-4)}$$

$$Z\{f(n)\} = \frac{1}{6}\left\{\frac{z}{z+2}\right\} + \frac{11}{6}\left\{\frac{z}{z-4}\right\}$$

$$f(n) = \frac{1}{6}Z^{-1}\left\{\frac{z}{z+2}\right\} + \frac{11}{6}Z^{-1}\left\{\frac{z}{z-4}\right\}$$

$$Z^{-1}\left[\frac{2z^2+3z}{(z+2)(z-4)}\right] = \frac{1}{6}(-2)^n + \frac{11}{6}(4)^n$$

Example 2: Find the inverse of Z -transform of $\frac{10z}{z^2-3z+2}$

Solution: Let $F(z) = \frac{10z}{z^2-3z+2} = \frac{10z}{(z-1)(z-2)}$

$$\frac{F(z)}{z} = \frac{10}{(z-1)(z-2)} - - - - - - (1)$$

$$\frac{10}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2} - - - - - - - - (2)$$

Multiply by $(z-1)(z-2)$

$$10 = A(z-2) + B(z-1)$$

$Put\ \ z = 1, 10 = A(-1)$	$Put\ z = 2, 10 = B(1)$
$A = -10$	$B = 10$

(2) becomes, $\frac{10}{(z-1)(z-2)} = \frac{-10}{z-1} + \frac{10}{z-2}$

$$\frac{F(z)}{z} = \frac{-10}{z-1} + \frac{10}{z-2} \quad Using\ (1)$$

$$F(z) = \frac{-10z}{z-1} + \frac{10z}{z-2}$$

$$Z[f(n)] = -10\left(\frac{z}{z-1}\right) + 10\left(\frac{z}{z-2}\right)$$

$$f(n) = -10Z^{-1}\left[\frac{z}{z-1}\right] + 10Z^{-1}\left[\frac{z}{z-2}\right]$$

$$f(n) = -10(1^n) + 10(2^n)$$

$$f(n) = -10 + 10(2^n)$$

$$Z^{-1}\left[\frac{10z}{z^2-3z+2}\right] = 10(2^n - 1)$$

Example 3: Find $Z^{-1}\left[\frac{z(z^2-z+2)}{(z+1)(z-1)^2}\right]$ by method of partial fractions.

Solution: Let $F(z) = \frac{z(z^2-z+2)}{(z+1)(z-1)^2}$

$$\frac{F(z)}{z} = \frac{z^2-z+2}{(z+1)(z-1)^2} - - - - - (1)$$

$$\frac{z^2-z+2}{(z+1)(z-1)^2} = \frac{A}{z+1} + \frac{B}{z-1} + \frac{C}{(z-1)^2} - - - - - (2)$$

Multiply by $(z+1)(z-1)^2$

$$\frac{\{z^2-z+2\}(z+1)(z-1)^2}{(z+1)(z-1)^2}$$

$$= \frac{A(z+1)(z-1)^2}{z+1} + \frac{B(z+1)(z-1)^2}{z-1} + \frac{C\,(z+1)(z-1)^2}{(z-1)^2}$$

$$z^2-z+2 = A(z-1)^2 + B(z+1)(z-1) + C(z+1)$$

$Put\ \ z = 1$	$Put\ \ z = -1$
$1-1+2 = 2C$	$(-1)^2+1+2 = A(-2)^2$
$2 = 2C \Rightarrow C = 1$	$4 = A(4) \Rightarrow A = 1$

$$z^2-z+2 = A(z^2+1-2z) + B(z^2-z+z-1) + C(z+1)$$

$$Equating\ \ the\ \ coefficient\ \ of\ z^2, 1 = A + B$$

$$B = 1 - A = 1 - 1$$

$$B = 0$$

(2) becomes, $\frac{z^2-z+2}{(z+1)(z-1)^2} = \frac{1}{z+1} + \frac{0}{z-1} + \frac{1}{(z-1)^2}$

$$\frac{F(z)}{z} = \frac{1}{z+1} + 0 + \frac{1}{(z-1)^2} \quad \text{Using (1)}$$

$$F(z) = \frac{z}{z+1} + \frac{z}{(z-1)^2}$$

$$Z[f(n)] = \frac{z}{z+1} + \frac{z}{(z-1)^2}$$

$$f(n) = Z^{-1}\left\{\frac{z}{z+1}\right\} + Z^{-1}\left\{\frac{z}{(z-1)^2}\right\}$$

$$f(n) = (-1)^n + n$$

Example 4: Find $Z^{-1}\left[\frac{z^2}{(z+2)(z^2+4)}\right]$ by the method of partial fraction.

Solution:

$$\text{Let } F(z) = \frac{z^2}{(z+2)(z^2+4)}$$

$$\frac{F(z)}{z} = \frac{z}{(z+2)(z^2+4)} ------(1)$$

$$\frac{z}{(z+2)(z^2+4)} = \frac{A}{z+2} + \frac{Bz+C}{z^2+4} -----(2)$$

Multiply by $(z+2)(z^2+4)$,

$$\frac{z}{(z+2)(z^2+4)}(z+2)(z^2+4) = \frac{A(z+2)(z^2+4)}{z+2} + \frac{(Bz+C)(z+2)(z^2+4)}{z^2+4}$$

$$z = A(z^2+4) + (Bz+C)(z+2)$$

$Put\ z = -2$ $-2 = A\{(-2)^2+4\} + 0$ $-2 = 8A$ $A = \frac{-1}{4}$	$Put\ z = 0$ $0 = 4A + 2C$ $2C = -4A$ $C = -2\left(\frac{-1}{4}\right) \Rightarrow C = \frac{1}{2}$
$z = Az^2 + 4A + Bz^2 + Cz + 2Bz + 2C$ $Equating\ co-efficient\ of\ z^2, \quad 0 = A + B$ $B = -A$ $B = -\left(-\frac{1}{4}\right) = \frac{1}{4}$	

$$(2)\text{ becomes}, \frac{z}{(z+2)(z^2+4)} = \frac{\frac{-1}{4}}{z+2} + \frac{\frac{1}{4}z + \frac{1}{2}}{z^2+4}$$

$$\frac{F(z)}{z} = \frac{-1}{4}\left[\frac{1}{z+2}\right] + \frac{1}{4}\left[\frac{z}{z^2+4}\right] + \frac{1}{2}\left[\frac{1}{z^2+4}\right] Using\ (1)$$

$$F(z) = \frac{-1}{4}\left[\frac{z}{z+2}\right] + \frac{1}{4}\left[\frac{z^2}{z^2+4}\right] + \frac{1}{2}\left[\frac{z}{z^2+4}\right]$$

$$Z\{f(n)\} = \frac{-1}{4}\left[\frac{z}{z+2}\right] + \frac{1}{4}\left[\frac{z^2}{z^2+4}\right] + \frac{1}{2}\left[\frac{z}{z^2+4}\right]$$

$$f(n) = \frac{-1}{4}Z^{-1}\left[\frac{z}{z+2}\right] + \frac{1}{4}Z^{-1}\left[\frac{z^2}{z^2+4}\right] + \frac{1}{2}Z^{-1}\left[\frac{z}{z^2+4}\right]$$

$$= \frac{-1}{4}Z^{-1}\left[\frac{z}{z+2}\right] + \frac{1}{4}Z^{-1}\left[\frac{z^2}{z^2+2^2}\right] + \frac{1}{4}z^{-1}\left[\frac{2z}{z^2+2^2}\right]$$

$$f(n) = \frac{-1}{4}(-2)^n + \frac{1}{4}2^n cos\frac{n\pi}{2} + \frac{1}{4}2^n sin\frac{n\pi}{2}$$

$$\left[\text{Since}, Z^{-1}\left[\frac{z}{z-a}\right] = (-a)^n, Z^{-1}\left[\frac{z^2}{z^2+a^2}\right] = a^n cos\frac{n\pi}{2}, Z^{-1}\left[\frac{az}{z^2+a^2}\right] = a^n sin\frac{n\pi}{2}\right]$$

Example 5: Find $Z^{-1}\left[\frac{z^3+3z}{(z-1)^2(z^2+1)}\right]$ by method of partial fraction.

Solution:

$$\text{Let } F(z) = \frac{z^3+3z}{(z-1)^2(z^2+1)}$$

$$\frac{F(z)}{z} = \frac{z^2+3}{(z-1)^2(z^2+1)} ------(1)$$

$$\frac{z^2+3}{(z-1)^2(z^2+1)} = \frac{A}{(z-1)} + \frac{B}{(z-1)^2} + \frac{Cz+D}{z^2+1} -------(2)$$

Multiply by $(z-1)^2(z^2+1)$,

$$\frac{[z^2+3](z-1)^2(z^2+1)}{(z-1)^2(z^2+1)} = \frac{A(z-1)^2(z^2+1)}{(z-1)} + \frac{B(z-1)^2(z^2+1)}{(z-1)^2} + \frac{[Cz+D](z-1)^2(z^2+1)}{z^2+1}$$

$$z^2+3 = A(z-1)(z^2+1) + B(z^2+1) + [Cz+D](z-1)^2$$

$Put\ z=1$	$Put\ z=0, \quad 3=-A+B+D$
$4=2B$	$-A+D=3-B$
$B=2$	$-A+D=3-2$
	$-A+D=1---(3)$
$z^2+3=A(z^3+z-z^2-1)+B(z^2+1)+[Cz+D](z^2+1+2z)$	
$z^2+3=A(z^3+z-z^2-1)+B(z^2+1)+(Cz^3+Cz+2Cz^2)+D(z^2+1+2z)$	
$Equating\ coeff.\ of\ z^3$	$Equating\ coeff.\ of\ z$
$0=A+C------(4)$	$A+C+2D=0----(5)$
$Using\ (4) in\ (5), \quad 2D=0$	
$D=0$	
$Sub.\ D=0\ in\ (3)$	$Sub.\ A=-1\ in\ (4)$
$-A=1$	$-1+C=0$
$A=-1$	$C=1$

(2) becomes, $\frac{F(z)}{z}=\frac{-1}{(z-1)}+\frac{2}{(z-1)^2}+\frac{z+0}{z^2+1}$

$$F(z)=-\frac{z}{z-1}+\frac{2z}{(z-1)^2}+\frac{z^2}{z^2+1}$$

$$Z\{f(n)\}=-\frac{z}{z-1}+\frac{2z}{(z-1)^2}+\frac{z^2}{z^2+1}$$

$$f(n)=-Z^{-1}\left[\frac{z}{z-1}\right]+2Z^{-1}\left[\frac{z}{(z-1)^2}\right]+Z^{-1}\left[\frac{z^2}{z^2+1}\right]$$

$$Z^{-1}\left[\frac{z^2+3z}{(z-1)^2(z^2+1)}\right]=-1+2n+cos\frac{n\pi}{2}$$

Example 6: Find $Z^{-1}\left[\frac{z-4}{z^2+5z+6}\right](\text{Or})Z^{-1}\left[\frac{z-4}{(z+2)(z+3)}\right]$ by method of partial fraction.

Solution: Let $F(z)=\frac{z-4}{z^2+5z+6}$

$$F(z)=\frac{z-4}{(z+2)(z+3)}-----(1)$$

$$\frac{z-4}{(z+2)(z+3)}=\frac{A}{z+2}+\frac{B}{z+3}-----(2)$$

Multiply by, $(z+2)(z+3)$

$$\frac{z-4}{(z+2)(z+3)}(z+2)(z+3)=\frac{A(z+2)(z+3)}{z+2}+\frac{B(z+2)(z+3)}{z+3}$$

$$z-4=A(z+3)+B(z+2)$$

$Put\ z=-3$	$Put\ z=-2$
$-7=B(-3+2)$	$-6=A(-2+3)$
$-7=-B\Rightarrow B=7$	$-6=A\Rightarrow A=-6$

(2) becomes, $\frac{z-4}{(z+2)(z+3)}=\frac{-6}{z+2}+\frac{7}{z+3}$

$$F(z)=\frac{-6}{z+2}+\frac{7}{z+3} \quad Using\ (1)$$

$$Z\{f(n)\}=\frac{-6}{z+2}+\frac{7}{z+3}$$

$$f(n)=-6Z^{-1}\left(\frac{1}{z+2}\right)+7Z^{-1}\left(\frac{1}{z+3}\right)$$

$$f(n)=-6(-2)^{n-1}+7(-3)^{n-1}$$

Example 7: Find the inverse Z-transform of $\frac{z(z+1)}{(z-1)^3}$

Solution: Let $F(z)=\frac{z(z+1)}{(z-1)^3}$

$$\frac{F(z)}{z}=\frac{z+1}{(z-1)^3}=\frac{z-1+2}{(z-1)^3}[\because 1=2-1]$$

$$=\frac{(z-1)}{(z-1)^3}+\frac{2}{(z-1)^3}$$

$$\frac{F(z)}{z}=\frac{1}{(z-1)^2}+\frac{2}{(z-1)^3}$$

$$F(z) = \frac{z}{(z-1)^2} + \frac{2z}{(z-1)^3}$$

$$Z[f(n)] = \frac{z}{(z-1)^2} + \frac{2z}{(z-1)^3}$$

$$f(n) = Z^{-1}\left[\frac{z}{(z-1)^2}\right] + Z^{-1}\left[\frac{2z}{(z-1)^3}\right]$$

$f(n) = n + n(n-1)$ $\left[\text{Since, } \; Z^{-1}\left[\frac{z}{(z-1)^2}\right] = n; Z^{-1}\left[\frac{2z}{(z-1)^3}\right] = n(n-1)\right]$

$= n + n^2 - 1$

$f(n) = n$

Problems for Practise

1. Evaluate $Z^{-1}\left[\frac{z}{z^2+7z+10}\right]$ by method of partial fraction. Ans: $\frac{1}{3}(-2)^n - \frac{1}{3}(-5)^n$
2. Find $Z^{-1}\left[\frac{z}{(z+1)(z-1)^2}\right]$ by method of partial fraction. Ans: $\frac{-1}{4}(1)^n + \frac{1}{2}(n) + \frac{1}{4}(-1)^n$
3. Find $Z^{-1}\left[\frac{z^2+3z}{(z+2)(z-1)(z-5)}\right]$ by method of partial fraction. Ans: $\frac{-1}{2!}(-2)^n - \frac{1}{3}1^n + \frac{2}{7}5^n$
